Inhwa Song • Academic Statement of Purpose

University of Michigan, Computer Science & Engineering, Ph.D.
inhwa.song@kaist.ac.kr
https://inhwasong.com

Many challenges at both individual and societal levels stem from the limitations of personal worldviews and perspectives. I have witnessed how such limitations can contribute to prejudice and division through my experience of serving as a student president during the COVID crisis. In addition, my two years of experience as a developer and content moderator for my school's official online community provided firsthand insight into how the collective (or conflicting) worldviews of those involved in the development process can influence technology and, ultimately, the people who use them. Together, these experiences have guided my interest in how the interaction of diverse worldviews is both mediated by sociotechnical systems and also shapes technology development. Through my research, I aim to embrace pluralistic worldviews harmonically coexist, both through and within (socio)technology. For my Ph.D., I am excited to examine research questions at the intersection of human-computer interaction(HCI), social computing, and natural language processing(NLP). To this end, I hope to pursue my doctoral studies at the University of Michigan, Computer Science & Engineering.

My research interest has both guided and been shaped by my involvement in many exciting research projects across diverse academic domains and industry settings. While training as a peer counselor, I witnessed firsthand how individuals project and shape their worldviews during personal challenges. Some shared their experiences of using LLM chatbots, highlighting both potential and risks. Motivated by these insights, I seized the opportunity to collaborate with **Prof. Munmun De Choudhury** during my exchange period at Georgia Tech. I led timely research to examine how individuals understand and use LLM chatbots when seeking support for their mental health needs in real-world settings [2]. I conducted semi-structured interviews with 21 participants from every inhabited continent by connecting with local forums and support groups. Grounding analysis in psychotherapy literature of what makes support effective, we introduced the concept of therapeutic alignment. Our work has been successfully submitted to *CSCW'24* with me as the first author, currently undergoing review following a request for revisions. I realized that providing interaction components for the user to gain control within a reliable structure is the key design consideration of building responsible generative AI systems, further motivating me to not only envision but to build and test the reliability of such systems.

This motivation led me to a research internship at NAVER AI Lab to work with **Dr. Young-Ho Kim**, whom I have long admired for his novel system contributions to HCI research. I developed ExploreSelf, an LLM-driven interactive system that supports users in exploring their narratives of personal challenges by actively engaging with adaptive guidance [1], grounding in a formative study. Even with my vast hands-on experience managing the development and deployment of complex computing systems, this particular work was especially valuable to me for three main reasons. First, I gained a deeper understanding of what system contribution means in HCI research. Second, this research I led received a *Revise & Resubmit decision in CHI'25* with compliments for the high research quality. Last but not least, I recognized my ability to quickly learn and apply new technologies and programming methods, along with the confidence that I can effectively tackle any new technical challenges I face in future research.

During my internship at NAVER AI Lab, I could communicate with AI researchers from diverse backgrounds, each bringing a different worldview. For instance, within the goal of responsible AI, stakeholders' perspectives diverged between creating safety benchmarks, embedding safety in LLMs, and investigating what responsible AI means in the context of Korean society. Through direct conversations with stakeholders, I saw firsthand how such

divergent views stemmed from the tension between qualitative societal contexts and the quantitative methods that dominate ML. These discussions reinforced my commitment to bridge the communication gap between diverse worldviews. They also motivated me to broaden my own perspective on navigating the balance between technical feasibility and social requirements. While I have experienced translating qualitative insights into concrete system design [2], I also wanted to tackle research questions that involve direct intervention in language models. Under the guidance of **Prof. Alice Oh**, my ongoing project involves constructing a dialogue dataset of unstructured learner inputs paired with concise queries by fine-tuning the Phi-3 model with the StackExchange dataset, where each post combines a body of long-form explanations and a title of a focused question.

These research projects gave me the opportunity to adopt a wide range of approaches, from system building and evaluation [1,8], large-scale data analysis [4], fine-tuning language models, to qualitative methods such as semi-structured interviews [2] and cultural probe [3], and also systematic reviews [7]. Above all, I have developed my ability to identify valuable research questions and learned how to transform an initial idea into systematic research.

My primary research objective lies at the intersection of HCI, social computing, and NLP, with the ultimate aim of enabling technologies to foster and adapt to the diverse worldviews that users bring. For example, I am particularly interested in examining the design of interaction components that provide users with autonomy to tailor their interactions with AI systems to their unique backgrounds and contexts. From my experience of developing ExploreSelf [1], which involved offering multiple layers of support in the navigation of personal narrative, I learned that achieving user autonomy in human-AI interaction requires not only providing immediate control but also fostering self-awareness about how their interactions shape and are shaped by AI systems. Moving forward, I hope to investigate questions such as how AI systems can balance individual autonomy and collective alignment in multi-user/agent collaborative settings, providing a responsible environment for diverse perspectives to interact.

With my focus and potential directions to keep in mind, Michigan CSE is the ideal place to pursue my doctoral studies. In particular, I hope to work with **Prof. Q. Vera Liao**, whose expertise in embedding trust and control in AI systems, will provide a rich context to my proposed research. Also, it was her Explainable AI tutorials across HCI and AI conference venues that primarily inspired me to broaden my academic perspective by engaging with interdisciplinary communities, such as presenting my CSCW research [2] to the ACL audience. I aspire to continue bridging the communication gap between disciplines throughout my academic career. I am also eager to receive the tutelage of **Prof. Farnaz Jahanbakhsh**, whose work on designing and building social computing systems that enhance autonomy closely aligns with my interests. My experience in developing and deploying learnersourcing systems for real-class use and serving as the developer of my university's online community equips me to contribute meaningfully and foster strong synergy in our collaboration. Furthermore, I am drawn to the work on **Prof. Eytan Adar**, whose research has continuously provided me with insights into designing novel human-AI interaction components.

Upon completing my Ph.D., I aspire to remain in academia as a professor, continuing to research and mentor students to advance my vision of embracing the coexistence of diverse worldviews, both through and within technology. With my rich research experience, commitment to broadening my own perspective, and dedication to fostering societal harmony, I am confident in my ability to succeed in the University of Michigan CSE Ph.D. program and contribute meaningfully to their mission to build a future that will improve the lives of all people.

References:

[1] **Inhwa Song**, SoHyun Park, Sachin R. Pendse, Jessica L. Schleider, Munmun De Choudhury, Young-Ho Kim. ExploreSelf: Fostering User-driven Exploration and Reflection on Personal Challenges with Adaptive Guidance by Large Language Models.

1-31. *CHI 2025 (under review with revise & resubmit decision)*.

[2] **Inhwa Song***, Sachin R. Pendse*, Neha Kumar, Munmun De Choudhury. The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support. 1-21. *CSCW 2024 (under review with major revision decision)*.

[3] **Inhwa Song**, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong. Understanding God-Saeng Phenomenon: Young Generation's Pursuit of Authentic Well-being. 1-17. *(in prep for submission to DIS 2025, submitted but not accepted to DIS 2024)*.

[4] **Inhwa Song***, David Gomez*, Ray Hung*, Charles Nimo*, (2023) Deep Learning for Identifying Language Features that Differentiate Mental Health Communities on Reddit. 1-8. *unpublished*.

[5] **Inhwa Song**, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong (2024) Bridging Values and Technology in God-Saeng Phenomenon: Designing AI for Self-Directed Well-being. 1-3. *CHI 2024 Workshop: Designing (with) AI For Wellbeing (accepted)*.

[6] Janghee Cho, **Inhwa Song**, Zainab Agha, Bengisu Cagiltay, Veena Calambur, Minjin "MJ" Rheu, Jina Huh-Yoo, Mobile Technology and Teens: Understanding the Changing Needs of Sociocultural and Technical Landscape. 1-11. *CHI 2025 Workshop Proposal (conditionally accepted)*.

[7] Siddhesh Pawar*, Junyeong Park*, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav, **Inhwa Song**, Alice Oh, Isabelle Augenstein (2024) Survey of Cultural Awareness in Language Models: Text and Beyond. 1-87. *Journal of Computational Linguistics (under review)*.

[8] Haesoo Kim, **Inhwa Song**, Juho Kim. (2022) Learnersourcing Modular and Dynamic Multiple Choice Questions. 1-5. *L@S 2022 Workshop on Learnersourcing Student Generated Content (accepted)*.